

discovered by them of course become very different, falling in different sub-spaces.

Now let us move on to other questions from last month's column. As you know, one of the biggest challenges in applying deep learning techniques to any classification is the need for huge amounts of training data. Let us assume that you have been asked to classify each sentence in a large corpus of sentences as positive, negative or neutral, which is the well known sentiment classification task. Now you have the choice of using a standard machine learning technique such as Support Vector Machines using a Bag of Words feature vector. This definitely seems simple and straightforward.

Let us also assume that given your recent interest in deep learning, you want to see if you can use a neural network such as a feed forward neural network. In that case, the first question you need to answer is whether you have adequate data to train your neural network. You have two choices: (a) You can either go with a simple feed forward neural network with two hidden layers of dimension X or, (b) you can opt for a deep forward neural network with five hidden layers of dimension X . Would the same labelled data set size suffice for both cases? As you may have guessed, the deeper the neural network, the more labelled data it needs for training it without overfitting. While there are different techniques such as drop-out, shared weights, etc, which avoid over-fitting, one of the simplest ways to do this is to have adequate amounts of training data. Since you have been given a set of labelled data samples, in last month's column readers were asked to come up with techniques which could help increase the size of labelled data.

If the input data is in the form of images, one of the simplest ways of increasing the labelled data set's size is by applying label-invariant transformations such as scaling, rotation, etc, to obtain a new image to which we can apply the same label. If you are instead given a labelled set of sentences for sentiment classification, as in our example above, how would you increase the size of the training data? Well, one of our readers had pointed out that we could use a lexicon based approach to create new labelled samples, wherein we can replace individual words with their equivalent words from the lexicon. This is a good answer. Can you think of other ways by which you can generate multiple different sentences from the same sentence, without changing its target sentiment?

By now, the attentive reader would have figured out that we did not really answer the question on how much training data would suffice for training a neural network without overfitting. Well, there is no single right answer in this case. It would depend on how many parameters

you have in the network, which in turn depends on the number of hidden layers and the dimensions of each layer. One often-quoted number is 100,000 instances of labelled data distributed across all the categories that you are trying to classify. But please take this number with a pinch of salt. While a smaller number may suffice for a shallower network, you may need a larger number of labelled instances for a much deeper network. I have not found any good rule which explains, quantitatively, how much training data would be sufficient, given a fixed deep neural network architecture. Also, you need to think about the quality of your labelled data. For instance, let us assume that you are doing sentiment classification and you have the following two sentences:

- (a) This movie was awfully bad, it was a waste of money.
- (b) This restaurant was awfully bad, just waste of money.

Though these two sentences may be two separate labelled instances, the second sentence does not allow the model to learn anything new as it is very similar to the first sentence. Hence, it is not just the quantity of the labelled data, but also the quality that matters when deciding the performance of your approach. Ultimately, it is the performance of your algorithm on the test data that matters. Hence, one way of making sure that your network is not overfitting is to measure the training error and test error on your data set. If you find that by increasing the training data size, your test error reduces, the model is learning new information from the augmented training data, which allows it to generalise well on the unseen test data. On the other hand, if you find that only the training error reduces, but the augmented training data does not improve the test error, you are better off without augmenting the training data. There is a detailed discussion on how much training data is sufficient for training a deep neural net on Quora, which you may want to look at: <https://www.quora.com/How-much-data-is-enough-to-train-a-deep-NN-model>.

If you have any favourite programming questions/ software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, wishing all our readers a wonderful and productive month ahead! 

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